

MBAI5310G AI Programming

Assignment 5

Unsupervised Learning for Pattern Discovery and Customer Segmentation

ReturnWise Retail Analytics

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1. Understanding the Business Problem

The Main Business Problem

ReturnWise Retail Analytics is a decision-support project for a mid-sized online retailer that sells apparel, footwear, electronics, beauty products, home goods, and accessories. The company receives thousands of orders each month across multiple channels including its website, mobile app, marketplace partners, and social commerce platforms.

The main business problem is that product return rates are too high and the company does not have a clear picture of which types of orders or customers are driving those returns. Returned orders cost the company money in reverse shipping, inspection, restocking, refunds, and customer service time. They also make inventory planning harder because a product that looks sold may come back weeks later. The business needs to understand the patterns behind returns so it can take action before a return happens rather than after.

Why Customer Segmentation Is Useful

Not all customers and orders behave the same way. Some customers buy frequently and almost never return anything. Others consistently return products, possibly because of sizing issues, unmet expectations, or heavy discount-driven purchases. Segmentation helps the business move away from treating every order the same and instead identify distinct groups with different return risk profiles.

By grouping orders into meaningful segments, the business can understand which combinations of customer behavior, product type, discount level, and delivery experience are linked to returns. This gives the marketing and operations teams a much more targeted and efficient way to allocate their attention and resources.

Marketing Decisions That Can Improve

Discovering customer and order segments can improve several marketing and operational decisions. For high return risk segments, the business can proactively send better product guidance, sizing information, or delivery updates before the return window opens. For loyal low-return customers, the business can invest in retention campaigns, loyalty rewards, and personalized recommendations. For newer customers who have not yet established a return pattern, the business can focus on building a positive first experience through clear product information and reliable delivery.

2. Data Preparation

Dataset Overview

The dataset contains 455 order records with 22 columns. After removing 5 duplicate rows, the working dataset has 450 records. Missing values were found in four columns: Region had 7 missing entries filled with the most frequent region, Product_Price had 6 missing entries filled with the median price, Discount_Percent had 6 missing entries filled with the median discount, and Product_Rating had 7 missing entries filled with the median rating.

Feature Selection and Scaling

I selected 11 numerical features for clustering: Customer_Age, Product_Price, Discount_Percent, Delivery_Time_Days, Items_In_Order, Previous_Orders, Prior_Return_Count, Product_Rating, Product_Page_Views, Return_Window_Days, and

Returned. The Returned column was included because return behavior is the central signal this business cares about understanding. Identifier columns like Order_ID and Purchase_Date were excluded.

Before applying K-means, I scaled all features using StandardScaler. K-means works by measuring distances between data points, so features with larger numerical ranges would otherwise dominate the clustering. Scaling brings every feature to a mean of zero and a standard deviation of one so each feature contributes equally to the distance calculation.

3. Applying K-Means Clustering

Elbow Method

I used the Elbow Method to choose the number of clusters. This method trains K-means models for different values of K from 1 to 10 and records the inertia for each. Inertia measures how tightly packed the points within each cluster are. A lower inertia means the clusters are more compact. The chart below shows how inertia decreases as K increases.

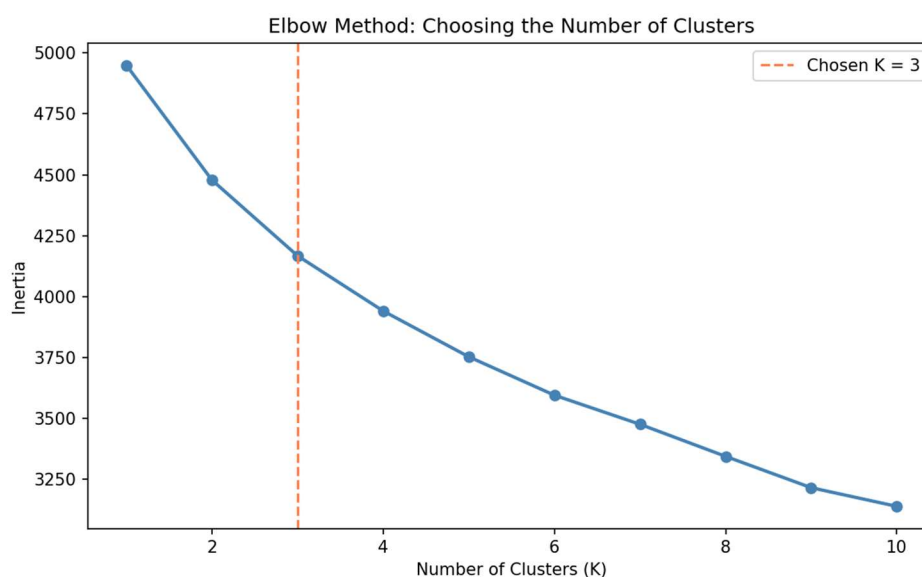


Figure 1: Elbow Method chart showing inertia across K values from 1 to 10

Choosing K = 3

Looking at the elbow chart, the inertia drops most noticeably between K=1 and K=3. After K=3 the reductions become smaller and more gradual, meaning that adding more clusters does not give a proportional improvement. Three clusters also produce results that make clear business sense for ReturnWise. The three groups that emerge are orders that were fully returned, orders from loyal frequent buyers with almost no returns, and orders from occasional safe buyers. These three segments are distinct from each other and directly actionable for the marketing and operations team.

Cluster Summary

After training the final K-means model with K=3, I analyzed the average feature values for each cluster. The table below summarizes the key characteristics of each segment.

Segment	Orders	Avg Age	Avg Price	Discount%	Prev Orders	Prior Returns	Rating	Return Rate
High Return Risk Orders	131	38.22	207.59	14.16	6.17	0.92	3.80	1.00
Loyal Frequent Buyers	116	35.06	210.08	10.04	15.32	1.72	4.00	0.03
Occasional Safe Buyers	203	40.20	218.33	10.74	3.65	0.51	4.01	0.00

4. Analyzing and Interpreting the Clusters

High Return Risk Orders

This segment contains 131 orders and every single one of them was returned, giving a 100 percent return rate. These orders received the highest average discount at 14.16 percent and had the lowest average product rating at 3.80. The customers in this group have fewer previous orders at 6.17 on average, suggesting they are not highly experienced with the brand. The pattern points to discount-driven purchases where customer expectations were not met, possibly due to product fit issues, misleading descriptions, or quality concerns.

Loyal Frequent Buyers

This segment contains 116 orders and has a return rate of only 3 percent. These customers have the highest number of previous orders at 15.32 on average, making them the most experienced and engaged group. Despite having the highest prior return count at 1.72 from their longer history with the brand, they are currently buying with much more confidence. Their product rating average is 4.00 and their discount level is the lowest at 10.04 percent, suggesting they are buying because they want the product rather than because of a promotion.

Occasional Safe Buyers

This is the largest segment with 203 orders and a return rate of zero. These customers have the fewest previous orders at 3.65 on average and the lowest prior return count at 0.51. Their return window is the longest at 40.97 days, yet they chose not to return anything. This suggests a group of careful buyers who research before purchasing and are satisfied with what they receive. They are newer or less frequent customers who have not yet built a strong relationship with the brand.

Most Valuable and Most Attention-Needed Segments

The Loyal Frequent Buyers segment is the most valuable because these customers bring consistent repeated revenue with very low return costs. They have already built trust in the brand and are the strongest candidates for loyalty programs and cross-selling campaigns.

The High Return Risk segment needs the most urgent attention. Every order in this group was returned, which represents a significant financial burden in reverse shipping, restocking, and refund costs. The connection between high discounts and 100 percent return rates suggests the discount strategy may be attracting purchases that do not align with customer expectations.

5. Cluster Visualizations

Scatter Plot: Previous Orders vs Product Price

I chose Previous_Orders and Product_Price as the two features for this scatter plot because both are continuous and spread across a wide range of values, which avoids the grid pattern that appears when integer-only columns are plotted against each other. The chart clearly shows the Loyal Frequent Buyers stretching further to the right because of their high previous order counts, while the High Return Risk and Occasional Safe Buyers cluster toward lower order counts.



Figure 2: Scatter plot showing customer segments by previous orders and product price

Bar Chart: Average Feature Values by Segment

This bar chart compares five key feature averages across all three segments side by side. It clearly shows that Loyal Frequent Buyers have by far the highest Previous_Orders count. The High Return Risk segment stands out with the highest Discount_Percent and the lowest Product_Rating. Occasional Safe Buyers are consistently moderate across all features. This comparison gives a clean visual summary of what makes each segment different.

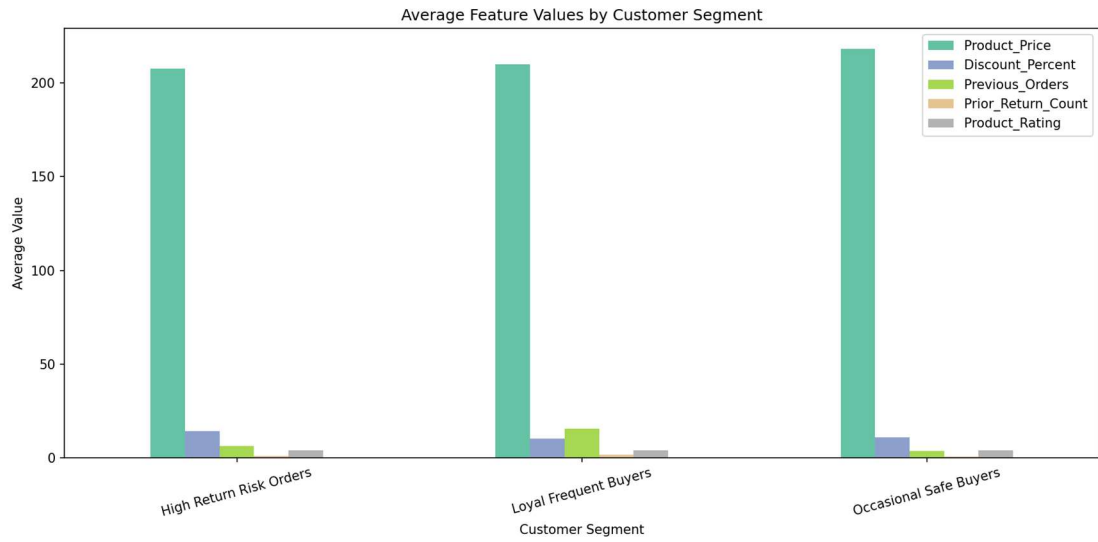


Figure 3: Bar chart comparing average feature values across the three customer segments

PCA Visualization

PCA reduces all 11 features into two components so the clusters can be visualized in a 2D chart. The two components together explain about 23.65 percent of the total variance in the data. The High Return Risk segment forms a distinct band that is clearly separated from the other two groups along the horizontal axis. The Loyal Frequent Buyers and Occasional Safe Buyers overlap somewhat in the middle, which makes sense since both groups have very low return rates. The PCA chart confirms that the three segments are genuinely distinct and not randomly assigned groups.

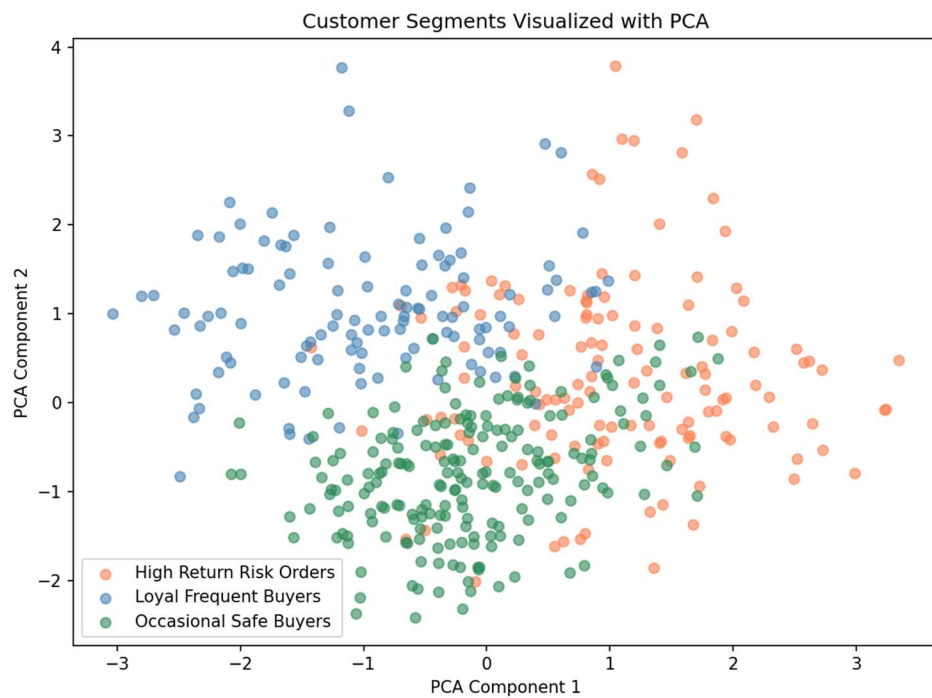


Figure 4: PCA scatter plot showing all three segments reduced to two dimensions

6. Business Recommendation

Patterns Discovered

The clustering revealed three clearly different order profiles. The most striking pattern is that orders with the highest discount levels and lower product ratings cluster together and all end up being returned. This tells me that discount-driven purchases are creating a return problem. The second pattern is that customers with a long purchase history show very low current return rates, even though they have more prior returns overall. The third pattern is that a large group of customers with little purchase history are making safe return-free purchases, possibly because they research carefully before buying.

How the Business Can Use These Segments?

The business can use these segments to move from reacting to returns after they happen toward managing return risk earlier. For the High Return Risk segment, the company can identify orders at checkout that match this profile and trigger automated product guidance emails, better size recommendation tools, or proactive customer support. For Loyal Frequent Buyers, the focus should be on retention and cross-selling since these customers already trust the brand. For Occasional Safe Buyers, the opportunity is to convert them into more frequent buyers by showing them the brand is reliable and worth returning to.

Marketing Strategy for Each Segment

For High Return Risk Orders, I would suggest reviewing the discount and promotion strategy to ensure it attracts customers who genuinely want the product rather than impulse buyers. Investing in better product page quality, clearer size guides, and more accurate product photos would also help reduce expectations mismatches.

For Loyal Frequent Buyers, I would suggest a loyalty program with tier-based rewards, personalized product recommendations based on purchase history, and early access to sales or new arrivals. These customers are already valuable and keeping them engaged is more cost-effective than acquiring new ones.

For Occasional Safe Buyers, I would suggest an educational content strategy that helps them discover more products they are likely to enjoy. Curated recommendations based on their previous safe purchases and gentle re-engagement emails after periods of inactivity could help grow this group into more frequent buyers.

How This Analysis Improves Decision-Making?

Before this analysis, the business was likely treating all orders the same and reacting to returns after they happened. The segmentation gives the team a way to prioritize where to invest marketing resources. Rather than spreading budget equally, the team can direct retention efforts toward loyal buyers and prevention efforts toward high risk orders where the return cost is already predictable. It also shifts the conversation from blaming customers for returns to understanding what operational or product factors are driving them.

7. Limitations and Responsible AI Reflection

Dataset Limitation

The dataset is synthetic and may not reflect real customer behavior accurately. Missing values in Region, Product_Price, and Product_Rating were filled with median or mode values, which may have introduced some noise into the clustering. In a real retail setting, the patterns would likely be messier, with more overlap between segments and more influence from external factors like seasonality, economic conditions, and product availability.

Why K-Means Does Not Always Produce Perfect Segments

K-means has several built-in limitations that affect how well it captures real customer groups. It requires the user to choose the number of clusters K before running, and that choice involves judgment rather than a definitive right answer. K-means also assumes that clusters are roughly spherical and similar in size, which real customer groups rarely are. The algorithm is sensitive to outliers because extreme values pull the cluster centers toward them. It will always find K groups even if the natural structure of the data does not support that many distinct segments.

Risk of Bias Without Human Review

If the business uses cluster labels to automatically apply stricter policies to customers in the High Return Risk segment, it could end up unfairly penalizing customers whose returns were caused by factors outside their control. A customer who returned a product because it arrived damaged, did not match the description, or had a sizing error in the company's own guide should not be treated the same way as someone exploiting the return policy. Applying automated restrictions based purely on a cluster label could damage brand trust and create ethical concerns.

Why Human Judgment Still Matters

Clustering is a pattern discovery tool, not a decision-making tool. It tells me what groups exist in the data based on numbers but it cannot tell me why a customer is in a particular group or what the right action is for them. A marketing manager reviewing a high-risk order needs to consider whether the product has known quality issues, whether the customer has an open complaint, and whether the return was caused by a business mistake rather than customer behavior. Using model results without human review also risks encoding historical patterns into future decisions. If a certain product category had poor descriptions in the past and generated many returns as a result, the model will learn that category is risky. A human reviewer would know to fix the descriptions rather than flag customers from that category as problematic. The model highlights where to look. The judgment about what to do belongs to the people who understand the customers and the context.